

Multi-type load forecasting model based on random forest and density clustering with the influence of noise and load patterns

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ABSTRACT

Load forecasting (LF) models are essential for various smart grid applications, and their accuracy heavily relies on the quality of input load data and load types. Previous LF studies have ignored noise loads due to tampering, transmission failures, etc., and have not considered the fusion of different types of loads, both of which have an impact on load forecasting accuracy. To address these issues, this study introduces a novel multi-type load forecasting model named MLF-RFDC, based on random forest and density clustering, that enjoys three-fold ideas: (1) it treats load data from each electrical activity as an independent data matrix, capturing variation patterns unique to each load type; (2) it identifies and corrects noisy entries in each data matrix using a low-rank clustering approach, highlighting noises as outliers and restoring them through latent factor analysis; and (3) it combines noise-free data matrices from all load types to enhance LF accuracy from an ensemble perspective. Extensive experiments conducted on ten benchmark datasets and three real-world load datasets demonstrate that our proposed algorithm outperforms 11 state-of-the-art models. Specifically, the performance results are remarkable: (1) the anomaly data detection accuracy is enhanced by up to 15.66%; (2) the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) for anomaly data recovery show significant improvements of tens of times; and (3) the MAE, RMSE, MAPE, and R-squared (R^2) for load forecasting are the most favorable.

1. Introduction

Load forecasting (LF) lays a vital foundation for smart-grid applications, including optimal scheduling and transport, unit maintenance, reduction of fringe power generation cost, to name but a few. Typically, an LF model operates over a unary time-series data which indicates the variation patterns of power consumption along the time horizon in a autoregressive fashion. In particular, at each time snapshot, the LF model strives to predict the current value of load (i.e., power consumption in a region of interest) based on the immediately preceding values. Machine learning models have excelled in the LF tasks including neural networks [1], time-series analysis [2], regression analysis [3], support vector machine [4] and fuzzy forecasting [5], thanks to their remarkable pattern induction capabilities.

Despite their excellent accuracy performance, the current LF models mainly posit a stochastic yet stationary process of the power consumption time-series, missing out two critical characteristics of the load in the wild. First, the load data can be multivariate and multi-typed. In

practice, as shown in the leftmost of Fig. 1, the power consumed by a diversity of electrical endeavours are composed of time-series with disparate variation patterns such as residential utilities and commercial and industrial activities. Blending these patterns by treating the load variation as a unary time-series is most likely to overlook rich information conveyed by each decomposition of the load time-series corresponding to a distinct electrical endeavour, leading to inferior LF performance.

Second, the real-world load data can be noisy. Unexpected events abound in the grid, such as human errors, collection failures of smart meters, network issues in data transmission and storage, and malicious attacks against load data [6], all of which incur inaccurate load data collection. Such noises distort the predictive results of LF models not only at the time snapshots they emerge, but also tend to escalate their negative impacts in the subsequent predictions due to the LF models' autoregressive nature. The more frequently the noisy load data are presented, the more largely the model predictions deviate to the true

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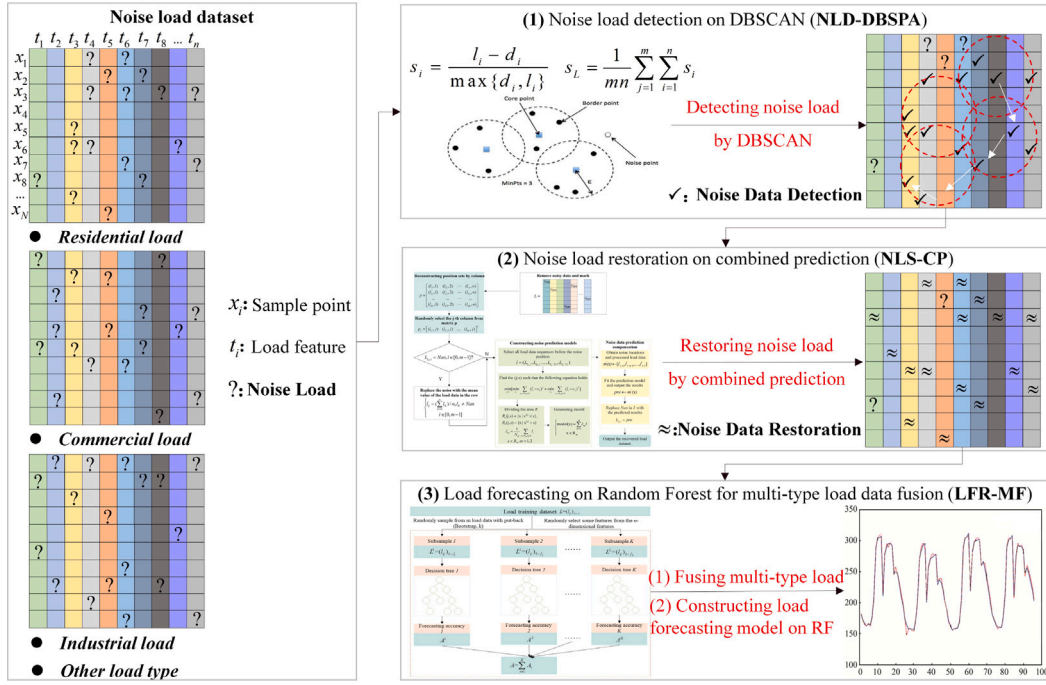


Fig. 1. Logic idea of MLF-RFDC.

load regression. Unfortunately, it remains under-explored to precisely locate the noisy inputs and repair their negative impacts on-the-fly in an LF regime.

However, the noise studied in existing load forecasting refers to random white noise, which usually has certain probability distribution properties, such as Gaussian distribution [7]. To cope with the two unique characteristics of real-world load data, we in this paper propose a novel LF model, termed *Multi-Load Forecasting based on Random Forest and Density Clustering* (MLF-RFDC), aiming to enable accurate LF with respect to the diverse patterns of multiple power-consuming endeavours and noises. The main idea of MLF-RFDC is shown in Fig. 1, and described as follows.

We treat the load data from each electrical endeavour as an independent load data matrix, of which the (i, j) -the entry indicates the load of location i at the time snapshot t . To detect the possibly noisy entry, we pursue a low-rank density-based clustering structure from each load matrix, such that the noises are spotlighted as outliers of all clusters. Then, we restore the detected noisy entries with latent factor analysis. A random forest model is thereafter built upon the restored data, with trees grown from each load data matrix independently via bagging and bootstrapping and the prediction results of all trees are ensemble to yield the final load forecasting results.

Main work of this paper are listed as follows.

- This is the first study to cast the LF task with noisy inputs as a low-rank load matrix recovery task. A novel noise detection and restoration algorithm leveraging density-based clustering is proposed to reduce the impact of diversified electricity consumption patterns on the accuracy of noise load detection. A load recovery algorithm based on combined prediction is proposed to optimize the trade-off between accuracy and efficiency of noise restoration.
- A random forest-based load forecasting algorithm under the fusion of multiple types of load data is proposed to reduce the dependency of load forecasting model on the types of power-consuming endeavours and improve the accuracy performance of LF tasks.
- A theoretical analysis shows that by harmonizing the multi-type load data matrices via bagging and bootstrapping, a tighter generalization error bound is manifested over an independent LF model.

- Extensive experiments are carried out on ten public benchmark dataset and three real-world load datasets, and the results demonstrate the superior performance of our proposed MLF-RFDC approach over 11 state-of-the-art rival models in terms of load noise detection, load noise restoration, and load forecasting accuracy.

The rest of this paper is organized as follows. The related works are reviewed in Section 2. Sections 3 and 4 focus on details of noise load detection, restoration algorithms and load forecasting model, respectively. Section 5 conducts the comparative experiments on 13 datasets and analyzes the results. Section 6 concludes this paper and highlights future work.

2. Related work

In this section, we analyze the advantages and disadvantages of various existing load forecasting models. Then we first introduce the random forest algorithm and its application in power systems. Finally, the corresponding discussion and our solution are given.

2.1. Load forecasting model

Load forecasting for power system is based on the spontaneous changes in power load and the influence laws of economic and meteorological factors. Through the analysis and research of historical data, it explores its internal development laws, and makes pre-estimation and prediction of power demand [8–10]. More and more intelligent algorithms are being applied in load forecasting, including but not limited to statistical methods, traditional time series analysis and artificial intelligence(AI) algorithms [11–13].

In recent years, with the rise of machine learning and deep neural network algorithms, artificial intelligence (AI) algorithms have also made more contributions to the research of load forecasting. Jalalifar et al. [14] applied deep learning to short term power load forecasting. To reduce the complex learning and training process of deep learning methods, KNN is applied before the training process and during the forecasting in [15]. And, in the process of model fitting, the appropriate parameters can improve the performance of the model. Jalali et al. [1]

first used a novel discrete grey wolf optimizer (GWO) algorithm for optimizing convolutional neural networks (CNN) hyperparameters in time series regression forecasting problems. Dai et al. [4] made some improvements on support vector machine (SVM), and proposed a hybrid model integrated with intelligent methods for feature selection and parameter optimization by adopting security particle swarm optimization (SecRPSO) to optimize parameters of the SVM model. As new algorithms continue to emerge, more scholars consider mixing multiple models to improve the forecasting precision. Liu et al. [16] proposed a hybrid short-term load forecasting method based on fuzzy combination weights and the optimal support vector machine. The mixed logistic regression (MLR) and long short-term memory (LSTM) algorithms were combined in [17] to forecast, respectively, the loads corresponding to the low and high-frequency components. Massaoudi et al. [18] even employed three models, namely, light gradient boosting machine (LGBM), extreme gradient boosting (XGB) and multilayer perceptron (MLP) as a stacked approach, coping with the stochastic variations of the load demand.

The research objects of load forecasting can be subdivided into residents, businesses and industries because of the large differences in power consumption behaviors and patterns. In [19], Cheng et al. proposed a day-ahead probabilistic residential load forecasting method for residential loads based on a novel deep learning model and micrometeorological data. Chitalia et al. [20] selected nine different recurrent neural network/convolutional neural networks (RNN/CNN) and their hybrid algorithms to formulated for load forecasting in various commercial buildings. For industrial customers, [21] utilized temporal convolutional networks (TCN) to extract the hidden information and took advantage of the state-of-the-art LGBM to forecast industrial customers' loads.

Comments: Presently, main work on load forecasting are focused on a certain type of specific objects, and are limited to considering only specific user power consumption patterns. And, the quality of load data also has a direct impact on the forecasting effect. Most studies do not concentrate on data processing, or simply use statistical methods to process. Meanwhile, the existing load forecasting models based on AI algorithms are less efficient for forecasting high-dimensional and massive load data.

2.2. Random forest

Compared with the above AI and machine learning algorithms, random forest (RF) [22] has the following advantages: (1) they can directly handle high-dimensional data; (2) they have high training efficiency; (3) the number of hyperparameters is small. Therefore, RF has been widely used in classification, regression and other tasks [23]. RF are frequently used as "black-box" models, as they generate reasonable predictions across a wide range of data while requiring little configuration [24]. With its algorithm advantages, RF has been widely used in various fields such as industry, medical health, ecosystem and so on [25]. RF model was applied in photovoltaic (PV) power generation to calculate the weight value and select factors with higher feature importance as input data for the model in [26]. In [27], Amato et al. tried to enhance machine learning classification based on random forest to manage massive data in order to provide smart eHealth services. Hakala et al. [28] also chose random forest classifiers in their system to explore the protein functions by the given input protein sequence. Then, RF is also applied into the research of power system [29]. Reddy et al. [30] designed a framework by the RF-based classifier to the accurate detection and classification of various single and multiple power quality (PQ) disturbances. In [31], Aprillia et al. used quantile regression random forest to analyze stochastic load behavior by a day-ahead prediction interval (PI). Alfieri et al. [32] used quantile regression forests and random forests to build probabilistic forecasts and four wavelet-based decompositions for this task. Lahouar et al. [33] utilized random forest to construct a model for an online

learning process, characterized by immunity to parameter variations and internal cross validation.

With the development of technology, single random forest no longer has great advantages. Wang et al. [34] developed a rule extraction method named improved RF-based rule extraction (IRFRE) method to derive accurate and interpretable classification rules. A hybrid model with the combination of RFs technique and the ant-lion optimizer (ALO) was proposed in [35], where ALO was used to optimize the hyper-parameters of the RFs model to improve its performance in terms of accuracy and computational time. Goehry et al. [36] adopted a bottom-up forecasting strategy by use of RF and sequential expert aggregation to forecast the system load. In [37], a random forest model coupled with ensemble empirical mode decomposition (EEMD-RF) is proposed, where the RF is applied for modeling the changes of different modes.

Comments: Although random forest is widely used in load forecasting, with the advantage of strong anti-noise ability, high integration accuracy, fast training speed, less computational memory and so on, most of the existing studies do not take into account the quality of input data. In the process of data acquisition, the possibility of data tampering cannot be ruled out. Moreover, the existing random forest-based load forecasting models do not consider the impact of electricity consumption load patterns on the precision of load forecasting.

2.3. Discussion and our idea

The existing load forecasting algorithms currently have the following problems: (1) the influence of noise and electricity consumption patterns on load forecasting accuracy is not considered; (2) it is difficult to balance the accuracy and efficiency of noise load restoration in load forecasting with noise.

To deal with the above issues, we propose corresponding algorithms from three aspects: (1) a noisy load detection algorithm based on DBSCAN with parameter adaptive selection; (2) a noisy load restoration algorithm based on combined prediction; (3) a random forest-based load forecasting algorithm with electricity consumption patterns.

3. Detection and restoration of noise load

3.1. Problem description

Noise in load data can be caused by unexpected events in the grid, collection failures of smart meters, network failures in load data transmission or storage, and various types of network attacks against load data [6]. Inaccurate load data bring great challenges to efficient and accurate load forecasting. Hence, before constructing an efficient and accurate load forecasting model, we first reconstruct the noisy load dataset and then use the density clustering to achieve high-precision detection of noisy loads, and finally compensate the noisy data at irregular locations based on the combined prediction to obtain a load dataset without noise.

Definition 1. Let $L = (L_i)_{i \in [1, m]}$, $L_i = \{l_{i1}, l_{i2}, \dots, l_{in}\}$ denotes the load datasets, where l_{ij} is the load value at the j -th moment of the i -day, $l_{i \max}^i$ is the maximum load peak of the i -day, and $l_{i \min}^i$ is the minimum load valley of the i -day. When l_{ij} satisfies (1) $l_{ij} = \text{null}$; (2) $\max(l_{ij}) > l_{i \max}^i$; (3) $\min(l_{ij}) < l_{i \min}^i$. We call L_i a noisy load.

When the load sequence L_i contains vacant values as well as load values that do not conform to the normal load range, it will inevitably bring great errors to the load forecasting of various tiers based on L_i . Eventually, it will affect the safe and stable operation of power production and dispatch. To construct a high-precision load forecasting model, this paper first detects and fixes the noise load accurately to eliminate the influence of noise load on load forecasting.

3.2. Noise load detection

The existing noise load detection algorithms mainly include human experience, load curve similarity comparison and state estimation. However, the noise load detection algorithm based on human experience uses a priori knowledge to subjectively determine whether there is noise load, which is inefficient and highly subjective. The noise load detection algorithm based on the similarity comparison of load curves cannot identify the noise load data with insignificant changes, while the noise load detection accuracy is low for the fusion of multiple types of load data, and the false detection rate and miss detection rate is high. To address these problems, this paper proposes the noise load detection algorithm based on DBSCAN with parameters adaptation (NLD-DBSPA).

Since the electricity consumption patterns are related to load types (e.g., residential, commercial, industrial) and seasons (e.g., spring, summer, autumn, winter), the overall characteristics are diverse and complex. To eliminate the impact of diverse and complex electricity consumption patterns on noisy load detection, this paper first reconstructs the noisy load data set by column with each day as a feature. And then, we use silhouette coefficient and noise ratio to detect noise load.

Definition 2. Suppose that there are k clusters $\{C_1, C_2, \dots, C_k\}$ in the load dataset L , d_i is the average distance from load point i in L to other load points in the same category, p_{ij} is the average distance from load point i in L to all load points in other categories, and let $p_i = \min\{p_{i1}, p_{i2}, \dots, p_{ik}\}$, then $s_i = \frac{p_i - d_i}{\max\{d_i, p_i\}}$ is called the silhouette coefficient of load point i .

Then the silhouette coefficient of load dataset L is $s_L = \frac{1}{mn} \sum_{j=1}^m s_i$, where $s_L \in [-1, 1]$.

Definition 3. Let N be the total number of noises in the load dataset L and N_d be the number of detected noises, then $N_c = \frac{N_d}{N}$ is called the noise ratio of the load dataset L .

In NLD-DBSPA, we first initialize the DBSCAN parameters ϵ and $Minpts$, perform clustering on the reconstructed noisy load dataset to obtain k clusters $C_1, C_2, \dots, C_k$, and then calculate the silhouette coefficients s_L and the noise ratio N_c of L until $|s_L - 1| \leq \sigma_1$ or $|1 - s_L| \leq \sigma_2$ and $|N_c - 1| \leq \sigma_3$. The NLD-DBSPA is shown in Algorithm 1.

3.3. Noise load restoration

The accuracy of noise load recovery directly affects the accuracy of the load forecasting model [38]. Existing noisy data recovery methods include split-box [39] and sparse matrix decomposition [40]. However, the accuracy of noisy data restoration based on split-box is low, and the time-consuming of noisy data restoration based on sparse matrix decomposition is high. Accordingly, we propose the noise load restoration algorithm based on combined prediction (NLS-CP).

Let $L = (l_{ij})_{m \times n}$ be the load data set and N be the set of noise data obtained based on noise load detection. We remove the noise data in N and mark it as Nan , and then obtain the noise positions in L and sort them by column.

For $\forall p_j = [(l_{1j}, j) \quad (l_{2j}, j) \quad \dots \quad (l_{mj}, j)]^T, j \in [1, n]$, if $L_{lij} = Nan, i \in [0, m-1]$ is satisfied, we recover the missing load values using the mean of all non-noise load data in the row where the noise location is located according to Eq. (1).

$$\begin{cases} l_{ij} = (\sum_{k=0}^n l_{ik})/n, l_{ik} \neq Nan \\ i \in [0, m-1] \end{cases} \quad (1)$$

Otherwise, all the load data sequences $l = (L_{l_{1j}}, L_{l_{2j}}, \dots, L_{l_{ij-2j}}, L_{l_{ij-1j}})$ before the noise location p_j are selected, and the noise load forecasting model $f(l_{ij})$ is constructed based on the regression binomial tree.

The flow of NLS-CP are shown in Fig. 2. And, Algorithm 2(NLS-CP) is described as follows.

Algorithm 1: NLD-DBSPA

Input: $L = (L_1, L_2, \dots, L_m), \epsilon, Minpts$
Output: Noise load dataset N

- 1 $L' \leftarrow ReconCol(L);$ //Reconstructing the load dataset by column
- 2 Marking all load points in L' with an unprocessed status;
- 3 $(s_L, N_c) \leftarrow Init(L, \epsilon, Minpts);$ //Initializing the Algorithm
- 4 **while** $(|s_L - 1| > \sigma_1 \text{ or } |1 - s_L| > \sigma_2) \text{ and } |N_c - 1| > \sigma_3$ **do**
- 5 **for each** p **in** L **do**
- 6 For object p , check the ϵ -neighborhood $N_{\epsilon}(p)$;
- 7 **if the number of objects contained in** $N_{\epsilon}(p)$ **is less than** $MinPts$ **then**
- 8 p is marked as a boundary point or noise point;
- 9 **else**
- 10 p is marked as a core point, and a new cluster C is created, and all points in the neighborhood of p are added to C ;
- 11 **for all** $q \notin N_{\epsilon}(p)$ **do**
- 12 **if** $N_{\epsilon}(p)$ **contains** $MinPts$ **objects then**
- 13 we add to C the objects in $N_{\epsilon}(p)$ that are not classified;
- 14 **for each** p **in** L **do**
- 15 **if** $p \in L$ **and** $p \notin C_i$ **then**
- 16 $N \leftarrow N.insert(p);$
- 17 **Return** N ;

Algorithm 2: NLS-CP

Input: $L = (L_1, L_2, \dots, L_m)$, Noise Locations Set $P_N = [p_1, p_2, \dots, p_N]$
Output: Recovered load dataset R_L

- 1 $P_N \leftarrow ReconCol(P_N);$ //Reconstructing position sets P_N by column
- 2 Mark all load points in L' .columns as unprocessed;
- 3 **for** $P_N[i]$ **in** P_N **do**
- 4 $L'[P_N[i]] = Nan;$
- 5 **for cols in** L' .columns **do**
- 6 $i = P_N[i][0], j = P_N[i][1];$
- 7 **if** $\exists L'_{ij} = Nan, i \in [0, n-1]$ **then**
- 8 $L'_{ij} = (\sum_{k=0}^m L'_{ik})/m, p_{ik} \neq Nan;$
- 9 **else**
- 10 $l = (L'_{i-n,j}, L'_{i-n-1,j}, \dots, L'_{i-2,j}, L'_{i-1,j});$
- 11 $l' = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\} \leftarrow l;$
- 12 Find the (j, s) such that the following equation holds
- 13 $\min[\min_{j \in J} \sum_{c_1} (l'_i - c_1)^2 + \min_{c_2} \sum_{x_i \in R_2(j,s)} (l'_i - c_2)^2];$
- 14 $(R_1(j, s), R_2(j, s)) \leftarrow Divid(R);$
- 15 $\hat{c}_m \leftarrow \frac{1}{N_m} \sum_{x_i \in R_m(j,s)} l'_i;$
- 16 $m(x) \leftarrow model(x) = \sum_{m=1}^M \hat{c}_m I;$
- 17 $m'(x) \leftarrow [l'_{i-n}, l'_{i-n-1}, \dots, l'_{i-1}];$
- 18 $pre \leftarrow m'(x);$
- 19 $L'_{ij} \leftarrow pre;$
- 20 $R_L \leftarrow L';$
- 21 **Return** R_L ;

4. LFR-MF

The construction of load forecasting models is related to the load types (including residential, commercial, industrial, etc.) as well as the seasons and holidays affecting the loads [41]. Due to the large variability of load profiles for different load types or different seasons and holidays under the same load type, it is difficult to accurately predict other types of loads with prediction models trained based on the same type of load data.

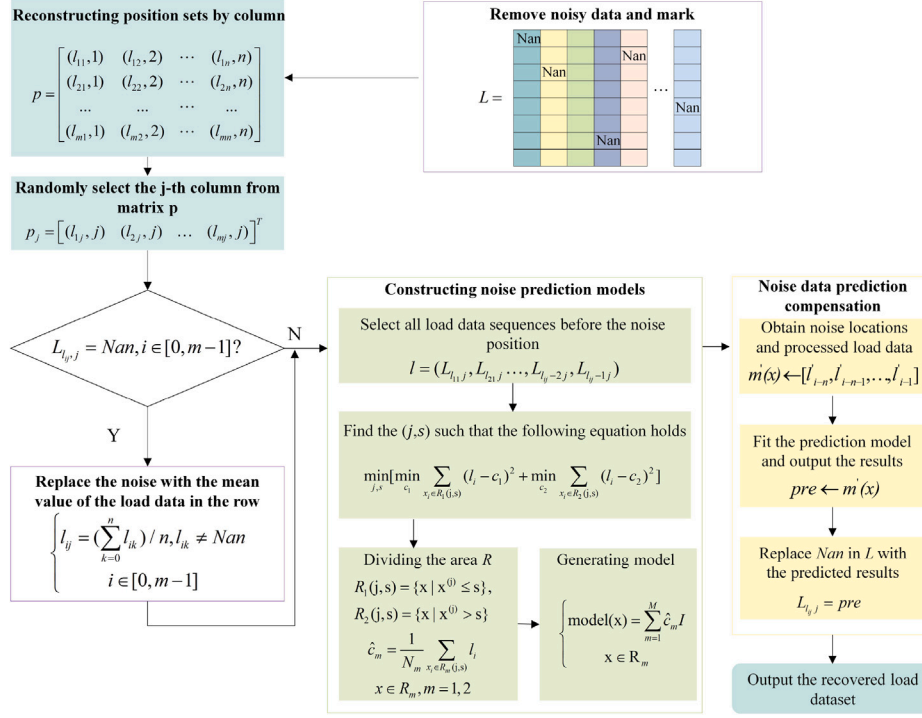


Fig. 2. The flow of NLS-CP.

Random forest is a typical ensemble learning regression algorithm. In the training phase, random forest uses bootstrap sampling to select several different sub-training datasets from the training dataset to sequentially build several different decision tree models. In the forecasting phase, the random forest averages the prediction results of multiple internal decision tree models to obtain the final prediction results [42]. Hence, to address the problem that it is difficult to accurately predict all load data by forecasting models only for a single type of load, we propose a load forecasting model mining algorithm based on random forest for multi-type load data fusion (LFR-MF).

4.1. Multi-type load data fusion

Let $L^R = (I_{ij}^R)_{m \times n}$ denote residential load, $L^C = (I_{ij}^C)_{m \times n}$ denote commercial load, and $L^I = (I_{ij}^I)_{m \times n}$ denote industrial load. For L^R , L^C and L^I , the corresponding load forecasting models are $f(I_{ij}^R)$, $f(I_{ij}^C)$ and $f(I_{ij}^I)$, respectively. From the perspective of forecasting, the accuracy of forecasting L^C and L^I based on $f(I_{ij}^R)$ is necessarily very low. This is because the electricity consumption patterns between L^R , L^C and L^I are inconsistent, and thus the forecasting model $f(I_{ij}^R)$ cannot cover the corresponding electricity consumption patterns of L^C and L^I . Consequently, by fusing different types of loads and introducing the concept of load correlation, we propose multi-type load fusion based on correlation coefficient (MLF-CC), which fuses load variables with low correlation among different types of loads, and finally can construct a dataset $L = \{L^R, L^C, L^I, \dots\}$ containing multiple types of loads.

Definition 4. Let $L^I = (I_{ij}^I)_{m \times n}$ and $L^C = (I_{ij}^C)_{m \times n}$ denote the two types of load matrices, respectively. $r_{L^I L^C} = \frac{n \sum_{j=1}^n I_{ij}^I I_{ij}^C - \sum_{j=1}^n I_{ij}^I \sum_{j=1}^n I_{ij}^C}{\sqrt{n \sum_{j=1}^n (I_{ij}^I)^2 - (\sum_{j=1}^n I_{ij}^I)^2} \sqrt{n \sum_{j=1}^n (I_{ij}^C)^2 - (\sum_{j=1}^n I_{ij}^C)^2}}$ is the Pearson correlation coefficient between any two sets of vectors in the L^I and L^C , and it satisfies $-1 < r_{L^I L^C} < 0$ or $0 < r_{L^I L^C} < 1$. Then $R = (r_{L^I L^C})_{m \times k}$ is said to be the correlation coefficient matrix of L^I and L^C .

By calculating the correlation coefficient $r_{L^I L^C}$ of the variables between different types of loads L^I and L^C , if $r_{L^I L^C}$ is closer to 0, it

means that the correlation between load variables L^I and L^C is lower. Then, we add both L^I and L^C to the new load data set L until all the load variables with lower correlations in different types of loads L^I and L^C are fused into L . The description of MLF-CC is shown in Algorithm 3.

Algorithm 3: MLF-CC

Input: Load dataset: $L_{m \times n}^I, L_{k \times n}^C$
Output: The fused load dataset L

```

1 for  $i \leftarrow 1$  to  $m$  do
2   for  $j \leftarrow 1$  to  $k$  do
3      $L_i^I \leftarrow (I_{i1}^I, I_{i2}^I, \dots, I_{in}^I)$ ;
4      $L_i^C \leftarrow (I_{i1}^C, I_{i2}^C, \dots, I_{in}^C)$ ;
5      $r_{L_i^I L_i^C} \leftarrow \frac{n \sum_{j=1}^n I_{ij}^I I_{ij}^C - \sum_{j=1}^n I_{ij}^I \sum_{j=1}^n I_{ij}^C}{\sqrt{n \sum_{j=1}^n (I_{ij}^I)^2 - (\sum_{j=1}^n I_{ij}^I)^2} \sqrt{n \sum_{j=1}^n (I_{ij}^C)^2 - (\sum_{j=1}^n I_{ij}^C)^2}}$ ;
6     if  $r_{L_i^I L_i^C} \approx 0$  then
7        $L \leftarrow L.insert(L_i^I, L_i^C)$ ;
8 Return  $L$ ;
```

4.2. LFR-MF

To ensure that the load forecasting model obtained from each decision tree contains as many load types as possible, in the LFR-MF, we randomly have put-back an equal amount of load data among m multi-type load data as the training dataset, and also randomly select a part of features among n -dimensional features to construct the decision tree. Thus, the correlation between individual decision trees in the generated random forest is small, and the accuracy of the load forecasting model is further improved.

In the process of constructing random forest, we build multiple decision trees by randomly selecting samples and features to reduce the correlation between decision trees and improve the accuracy of the model. The schematic diagram of the LFR-MF is given in Fig. 3.

Let the random forest $R_F = \{D_1, D_2, \dots, D_r\}$, where $D_i, i = 1, 2, \dots, r$ denotes the i th regression decision tree. For any D_i , the following property is satisfied.

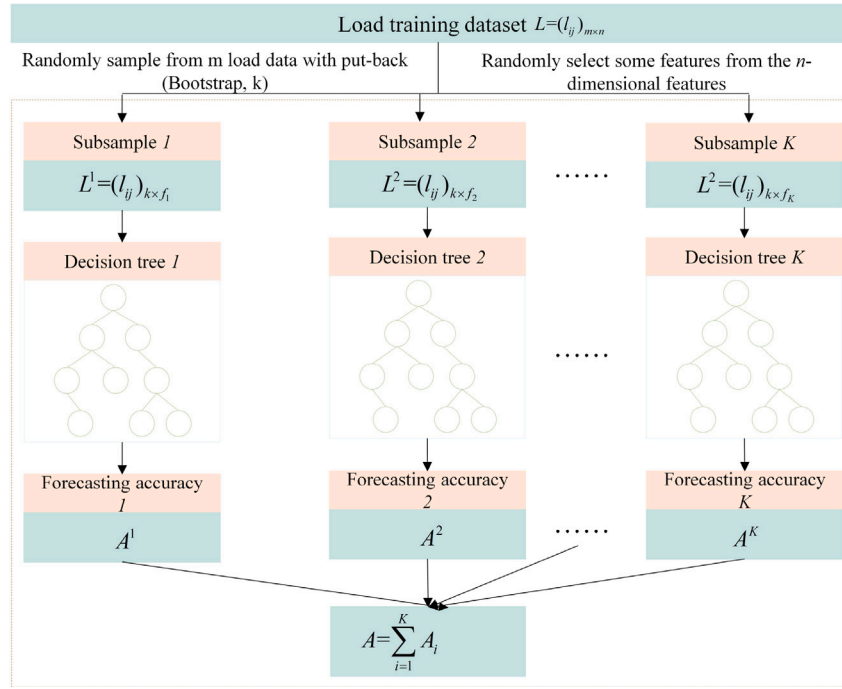


Fig. 3. The schematic diagram of the LFR-MF.

Property 1. From the load data set X , m samples are randomly selected with put-back as the training data set Z . For any sample in Z , let A denote that the sample is not selected in m selections, then $P(A) = (1 - \frac{1}{m})^m$.

Proof. Let B denote a sample p selected at random with a put-back from m samples. By probability theory, we know that $P(B) = \frac{1}{m}$. Then the probability that this sample p is not selected is $1 - \frac{1}{m}$. Therefore the probability that this sample p is not selected in m selections is $(1 - \frac{1}{m})^m$.

From Property 1, we can get that $\lim_{m \rightarrow \infty} P(A) = (1 - \frac{1}{m})^m = \frac{1}{e} \approx 0.368$. This indicates that the random and put-back selection of sample data makes each regression decision tree D_i contain almost all types of load data.

The LFR-MF is described as shown in Algorithm 4.

Algorithm 4: LFR-MF

Input: Load dataset: $L_{m \times n}^I, L_{k \times n}^C$; Radius of cluster: ε ; Density threshold: $Minpts$; Error: σ Number of regression decision trees: K

Output: Load Forecasting Accuracy A_c

```

1  $N^I \leftarrow NLD - DBSPA(L^I, \varepsilon, Minpts, \sigma)$ ;
2  $N^C \leftarrow NLD - DBSPA(L^C, \varepsilon, Minpts, \sigma)$ ;
3  $R_{N^I} \leftarrow NLS - CP(L^I, N^I)$ ;
4  $R_{N^C} \leftarrow NLS - CP(L^C, N^C)$ ;
5  $L \leftarrow MLF - CC(R_{N^I}, R_{N^C})$ ;
6  $\{L_1, L_2, \dots, L_m\} \leftarrow SamplePutBack(L, m)$ ;
7 for  $i \leftarrow 1$  to  $m$  do
8    $T_i \leftarrow RDTree(L_i, k)$ ;
9    $A_i \leftarrow CalAcc(T_i)$ ;
10   $sum \leftarrow sum + A_i$ ;
11  $A_c \leftarrow \frac{sum}{m}$ ;
12 Return  $A_c$ ;

```

From Algorithm 4, we know that the dataset L contains m samples, k features, and the time complexity of the multi-type load fusion is $O(mk \log m)$. Also, a weighted average of m regression decision trees is performed to obtain the overall prediction accuracy A_c . Thus, the time complexity of LFR-MF is $O(m(mk \log m))$.

4.3. Algorithm analysis

In this subsection, to test that the random forest can effectively handle datasets with multiple types of load and noise load, we analyze the generalization ability of the random forest algorithm. Breiman [22] pointed out that the randomness of samples and features makes the random forest have strong generalization ability. And its generalization ability can handle the efficient and accurate forecasting of load with multiple electricity consumption behaviors and noises.

Definition 5. Let X be the fused multi-type load training dataset and $T_1(X), T_2(X), \dots, T_k(X)$ be the k corresponding regression decision trees generated according to X . We randomly select samples in the load training data set distribution (X, Y) , where Y denotes the load values. And let Θ_k be a random vector with independent identical distribution and $T(X, \Theta_k) = T_k(X)$. Then, we call $m_r(X, Y)$ the margin function of the random forest, denoted as

$$m_r(X, Y) = a_{v_k}(T_k(X) = Y) - \max_{j \neq Y} a_{v_k}(T_k(X) = j) = P_{\Theta}(T(X, \Theta) = Y) - \max_{j \neq Y} P_{\Theta}(T(X, \Theta) = j) \quad (2)$$

The larger the value of $m_r(X, Y)$, the higher the confidence level of the classification result. And, let $r_{m_r}(\Theta, X, Y)$ be the original margin function of the random forest, denoted as $r_{m_r}(\Theta, X, Y) = I(T(X, \Theta) = Y) - I(T(X, \Theta) = \hat{j}(X, Y))$.

Definition 6. Let $m_r(X, Y)$ be the margin function of random forest, and we call $GE = P_{X,Y}(m_r(X, Y) < 0)$ the generalization error of the random forest.

Theorem 1. Let $m_r(X, Y)$ be the margin function of the random forest, Θ and Θ' be two independent and identically distributed random vectors, and $s = E_{X,Y}(m_r(X, Y))$. Also, let $\rho(\Theta, \Theta')$ be the correlation coefficient between Θ and Θ' , $\bar{\rho} = E_{\Theta, \Theta'}[\rho(\Theta, \Theta')]$ be the mean of the correlation coefficient $\rho(\Theta, \Theta')$. Then the generalization error of the random forest satisfies the following inequality

$$GE \leq \frac{\bar{\rho}(1 - s^2)}{s^2} \quad (3)$$

Proof. See Appendix A.

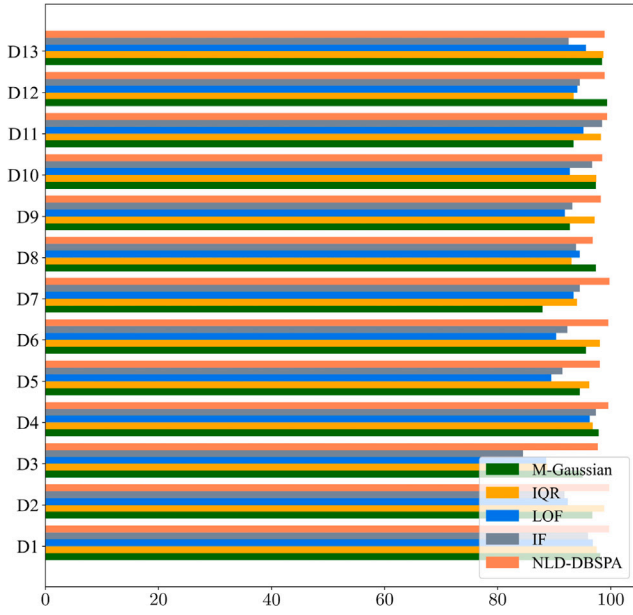


Fig. 4. Comparison of the precision between different noise load detection algorithms.

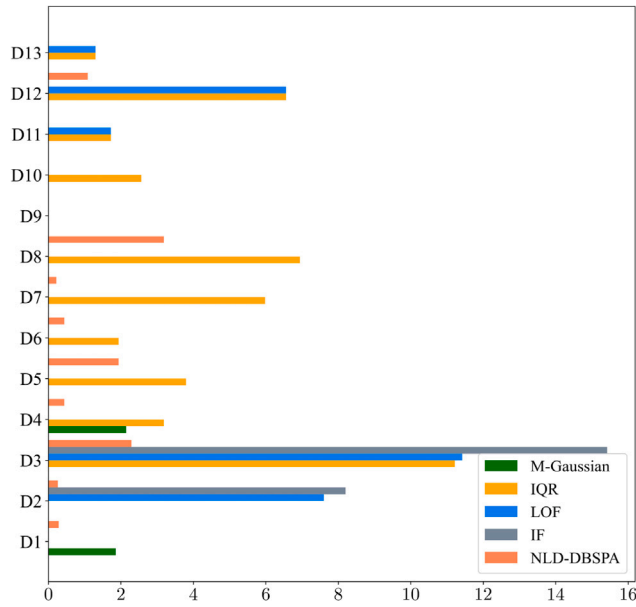


Fig. 5. Comparison of the false detection rate between different noise load detection algorithms.

In this paper, we construct regression random forests based on the input data (X, Y) and selectors, where each selector is used to delineate subintervals by fusing multiple types of load data. And $m_r(X, Y)$ is used as the boundary function of the subinterval to measure the degree of subinterval partitioning.

5. Experimental analysis

In this section, we conduct experiments on three real datasets and ten open load datasets to evaluate the performance of our proposed algorithms in this paper. First, we elaborate the experimental setting in terms of load data sources and features, evaluation metrics, and comparison algorithms. Second, we analyze the experimental results in terms of noisy data detection, noisy data repair, and load forecasting to

Table 1

Experimental datasets.

Abbreviations	Names	# Instances	# Features
D1	Nantong-Resident	159	96
D2	Nantong-Commerce	430	96
D3	Nantong-Mixed	159	96
D4	District1-Market	365	24
D5	District1-Resident	365	24
D6	District1-Restaurant	365	24
D7	District1-School	365	24
D8	District2-Resident	365	24
D9	District2-School	365	24
D10	District2-Warehouse	365	24
D11	District3-Hospital	365	24
D12	District3-Resident	365	24
D13	District3-Restaurant	365	24

Table 2

Experimental datasets with noise load data.

Abbreviations	Names	# Instances	# Features	# Noise
D1	Nantong-Resident	159	96	768
D2	Nantong-Commerce	430	96	2112
D3	Nantong-Mixed	159	96	768
D4	District1-Market	365	24	456
D5	District1-Resident	365	24	456
D6	District1-Restaurant	365	24	456
D7	District1-School	365	24	456
D8	District2-Resident	365	24	456
D9	District2-School	365	24	456
D10	District2-Warehouse	365	24	456
D11	District3-Hospital	365	24	456
D12	District3-Resident	365	24	456
D13	District3-Restaurant	365	24	456

show the advantages of the proposed algorithm in this paper. Finally, the authenticity and credibility of the experimental results are verified by Friedman test.

5.1. Data source

The 3 real load datasets are obtained from Nantong Power Supply Company of the State Grid Corporation of China. They can be downloaded at the . Their sizes include 430, 159 and 159 rows, respectively, and each day's load data is sampled at 15-minute intervals with a feature size of 96. 80% of each dataset is used as training data, and the remaining 20% is used as testing data. The 10 open load datasets are available from <https://data.openei.org/submissions/153>. They contain hourly load data for 16 commercial and residential properties in all TMY3 locations of the United States of America. The size of each load dataset is 365 and the number of features is 24. Similarly, 80% of each dataset is used as training data, and the remaining 20% is used as testing data. All experimental datasets are shown in Table 1.

To verify the performance and effectiveness of NLD-DBSPA and NLS-CP, for all the experimental datasets, we randomly inject different proportions of noise load data according to Eq. (4) so that the formed anomaly load dataset has exactly the same size and feature number as the original dataset [43]. Finally, we form the anomalous load datasets as shown in Table 2.

$$\tilde{x}(t) = \alpha(t) * load(t) = \begin{cases} \alpha = random(8, 10), t \in \{1-5, 9-10\} \\ \alpha = random(6, 8), t \in \{6-8, 11-12\} \end{cases} \quad (4)$$

5.2. Evaluation criteria

To evaluate the performance of the algorithms proposed in this paper, for all the experimental datasets, we use the precision, false detection rate, and the missing detection rate to evaluate the performance of various noisy data detection algorithms. We also use the average data recovery rate to evaluate the performance of various noisy

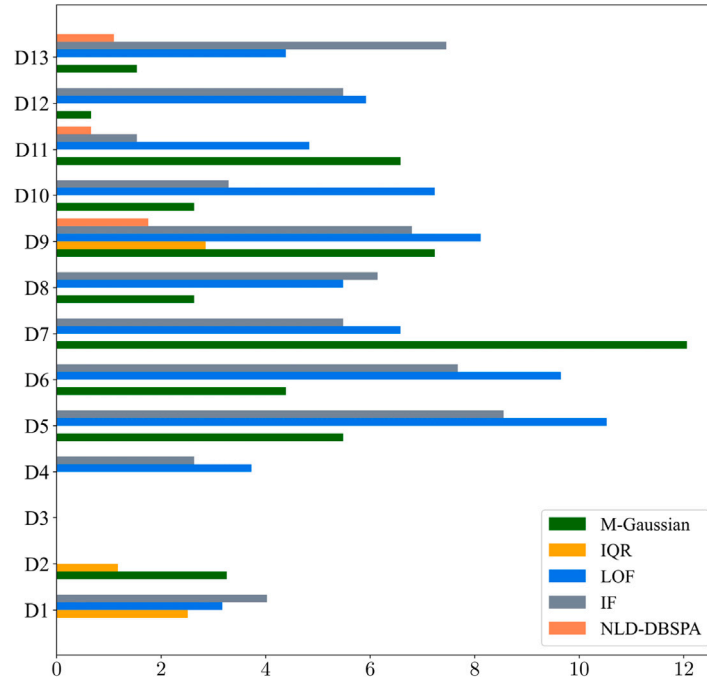


Fig. 6. Comparison of the missing detection rate between different noise load detection algorithms.

data recovery algorithms, and RMSE, MAE, MAPE, R^2 to evaluate the performance of various load forecasting models.

(1) Metrics for noisy data detection

To effectively evaluate the performance of various noise data detection algorithms, we set T_n to denote the number of normal data identified as normal data, F_n to denote the number of noisy data identified as normal data, T_a to denote the number of noisy data identified as noisy data, and F_a to denote the number of normal data identified as noisy data.

$P_r = \frac{T_n + T_a}{T_n + T_a + F_n + F_p}$ is called the precision of the noise data detection algorithm. The higher the P_r , the better the performance of the noise data detection algorithm.

$F_r = \frac{F_n}{T_n + F_n}$ is called the false detection rate of the noisy data detection algorithm. The smaller the F_r , the better the performance of the noisy data detection algorithm.

$M_r = \frac{F_a}{T_a + F_a}$ is called the missing detection rate of the noisy data detection algorithm. The smaller the, the better the performance of the noisy data detection algorithm.

(2) Metrics for noisy data restoration

Let $\bar{e}(d_r, d_t) = \sum_{i=1}^m \left| \frac{(d_t - d_r)}{d_t} \right| - 1/m$ denote the average data recovery rate of the noisy data restoration algorithm, where d_r is the recovered value of the load data at time t , d_t is the actual load value at time t , and m is the number of noisy data in each experiment.

(3) Metrics for load forecasting

And in this paper, we use RMSE, MAE, MAPE, and R^2 as shown in the following equations to evaluate the performance of the load forecasting model, respectively.

5.3. Baselines

We perform a comprehensive comparative study against our proposed model in this paper by considering various baselines as detailed below.

In this experiment, we perform a comprehensive comparative study in terms of noise load detection, noise load restoration, and load forecasting. In terms of noise load detection, NLD-DBSPA is compared with LOF [44], M-Gaussian [45], IQR [46], and Isolated Forest (IF) [47].

In terms of noise load restoration, NLS-CP is compared with the noise load restoration algorithm based on sparse matrix (NLS-SM) [40] and the noise load restoration algorithm based on mean imputation (NLS-MI) [48]. In terms of load forecasting, LFR-MF is compared with Linear Regression (LR) [49], XGBoost [50], Support Vector Regression (SVR) [51], DRN_STLF [52], and APLF [53].

5.4. Comparison and analysis

5.4.1. Performance comparison of noise load detection algorithms

For all the datasets, Fig. 4, 5 and 6 compare the precision, false detection rate and missed detection rate of different noise load detection algorithms, respectively. Also, to verify the difference of noise load detection performance by column and by row, we inject 50 noisy data into all datasets by Eq. (4), respectively, and the experimental results are shown in Fig. 7.

From these figures, we have the following conclusions:

- From Fig. 4, for D8 and D12, the precision of NLD-DBSPA is about 0.5% lower than that of M-Gaussian. While for the other datasets in Table 2, the precision of NLD-DBSPA is higher than that of all compared algorithms. Especially, for D3, the precision of our algorithm is improved by about 15.66% at maximum over IF.
- Although NLD-DBSPA has false detections on 10 noisy datasets from Fig. 5, the number of false detections is extremely small, with an average false detection rate of only about 0.8%.
- And, the results in Fig. 6 show that the missing detection rate of NLD-DBSPA is zero for 10 of the 13 load datasets, and the average missing detection rate on the remaining three datasets is about 1%. Compared with M-Gaussian, LOF, IQR, and IF, the average missing detection rate of NLD-DBSPA is reduced by 92.4%, 42.68%, 94.96%, and 96.81%, respectively.

From Fig. 7, we have the following conclusions:

- For all load datasets, the precision, false detection rate and missing detection rate of the noise load detection algorithm based on DBSCAN with parameters adaptation by column are significantly better than results of the noise load detection by row.

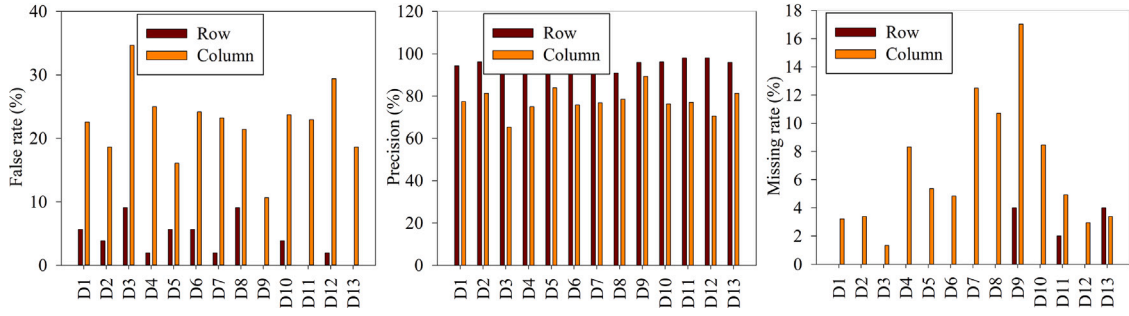


Fig. 7. Performance comparison of noise load detection based on DBSCAN with parameters adaptation by column and row respectively.

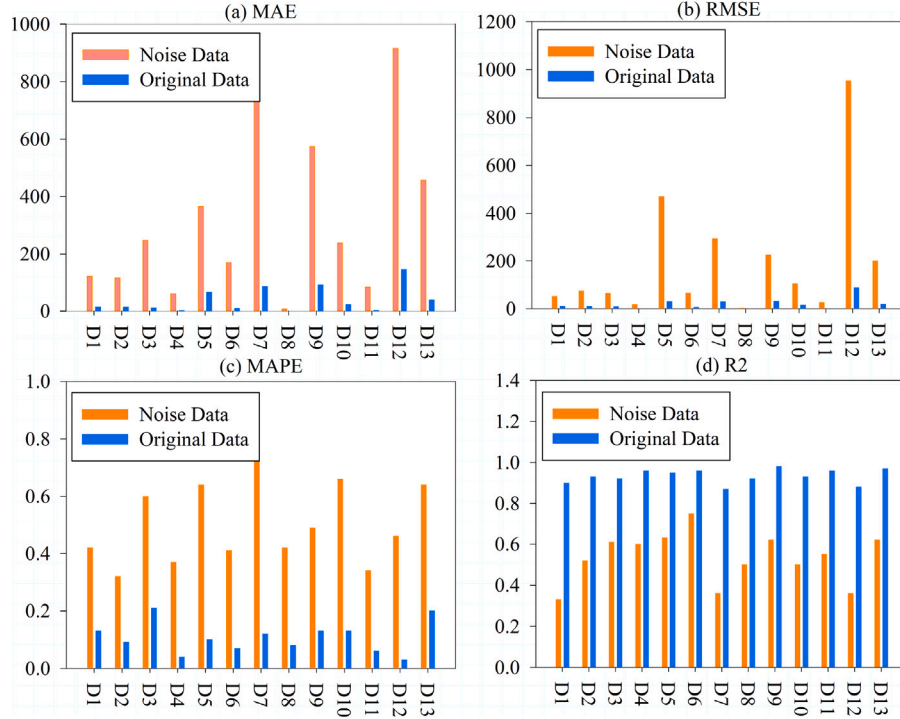


Fig. 8. Comparison of performance based on random forest for original load data and load data with noise.

- Specifically, for D9, noise load detection by column has the least improvement in precision over noise load detection by row, which is about 7.42%. For D3 and D12, the improvement in precision is the highest for noise load detection by column over by row, about 39.14% and 38.89%, respectively.
- These experimental results demonstrate that our proposed NLD-DBSPA has a greater advantage compared with other algorithms. This is because the same class of users has different electricity consumption behaviors at different moments, which affects the effect of noise detection. When we detect noise data by column, we only consider the electricity consumption behavior of the same class of users at the same moment in time. And when we detect noise data by rows, we need to consider the behavior of the same type of users at different times.

5.4.2. Performance comparison of noise load restoration algorithms

To verify the effect of noise on load forecasting, we construct load forecasting models based on random forest for all the data in Table 1 and Table 2, respectively. The statistical indicators of all datasets are shown in Fig. 8.

From Fig. 8, we have the following conclusions:

- MAE, RMSE, and MAPE on all datasets with noise are significantly higher than those on all corresponding original datasets. R^2 for all noisy datasets is significantly lower than R^2 for all original datasets.
- MAE on D4, D5, D6, and D11 with noise are about 1245.92%, 1469.41%, 1103.11%, and 1360.34% higher than MAE on the corresponding original datasets, respectively.
- RMSE on D5 and D11 with noise are significantly improved over RMSE on the original datasets by 3493.19%, 451.49%, respectively.
- MAPE on all datasets with noise was improved by 480.36% on average compared to MAPE on all original datasets. In particular, for D12 with noise, MAPE was improved the most with 1433.33%. Finally, the of all datasets with noise is significantly lower than that of the corresponding all original datasets, with a minimum decrease of 21.88% and a maximum decrease of 59.09%.

Overall, we can see from Fig. 8 that the MAE, RMSE, MAPE and of the random forest-based load forecasting model on the load datasets with noise are better than those on the original load datasets. We know that smaller MAE, RMSE, and MAPE represent better performance, and larger represents better performance. In general, the above experimental results show that constructing load forecasting model

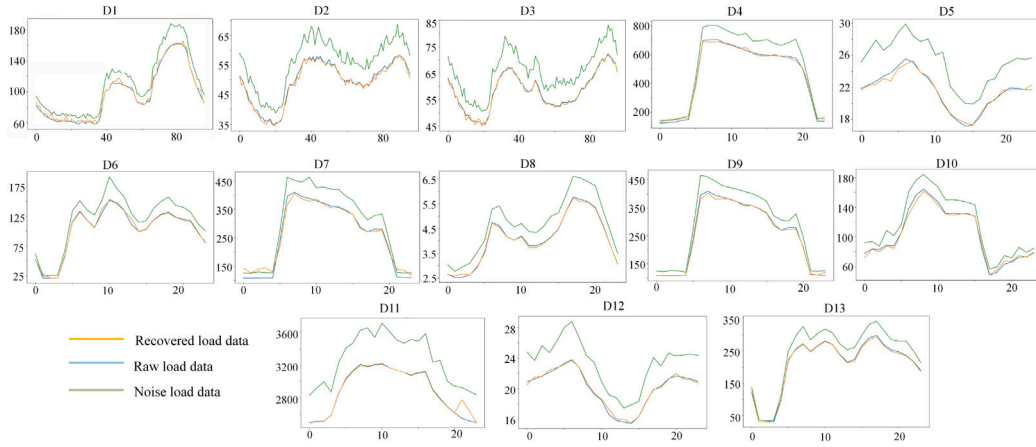


Fig. 9. Comparison of curves between original load data, restored load data and noisy load data.

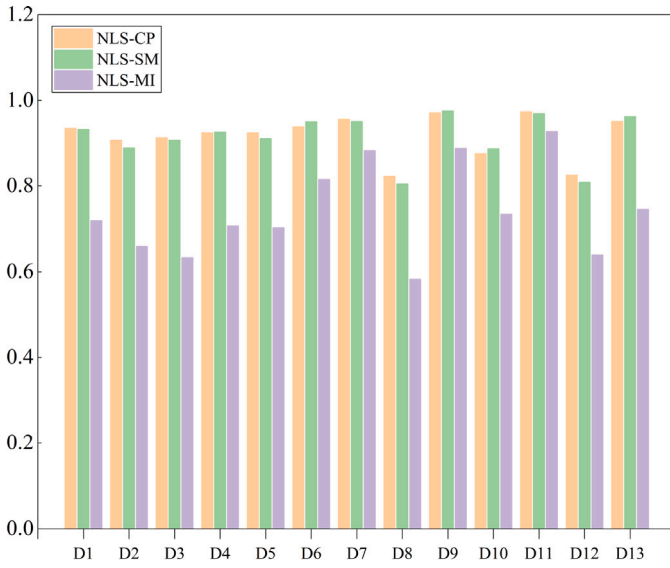


Fig. 10. Comparison of average data recovery rate between original load data, restored load data and noisy load data.

directly on load dataset with noise affects the electricity consumption patterns of various customers, which in turn affects the accuracy of load forecasting model.

To eliminate the influence of noisy load data on the precision, this paper uses a combined prediction method to repair the detected noisy load data. For the noisy load datasets in Table 2, Fig. 9 displays the load curves between before and after restoration of the noisy load data. And Fig. 10 compares the average data recovery rate between NLS-CP, NLS-SM and NLS-MI.

From these Figures, we have the following conclusions:

- From Fig. 9, for 13 noisy load datasets in Table 2, the noise load data curves recovered based on our proposed NLS-CP match very well with the original load data curves.
- From Fig. 10, average data recovery rate of the NLS-CP proposed in this paper is better than that of NLS-SM except for the D4, D6, D9, D10, and D13 datasets. And for the 13 loaded datasets with noise, the average data recovery rate of the NLS-CP is significantly better than that of the NLS-MI. Although for the D4, D6, D9, D10, and D13 datasets, the average data recovery rate of the NLS-CP are smaller than that of the NLS-SM by about 0.216%, 1.26%, 0.45%, 1.35%, and 1.19%, respectively.

Table 3

Comparison of the average time-consumption (Sec).

Abbreviations	NLS-CP (Our)	NLS-SM	NLS-MI
D1	2.6793 ^b	288.5478	0.6120 ^a
D2	2.0106 ^b	269.5668	0.6347 ^a
D3	2.1654 ^b	271.0145	0.6346 ^a
D4	0.5210 ^a	183.3664	0.5410 ^b
D5	0.5141 ^a	180.3022	0.5296 ^b
D6	0.4911 ^a	181.0412	0.5366 ^b
D7	0.4826 ^a	187.6854	0.5219 ^b
D8	0.5245 ^a	183.1129	0.5299 ^b
D9	0.5251 ^b	185.2399	0.5182 ^a
D10	0.4959 ^a	180.5223	0.5446 ^b
D11	0.5040 ^a	184.7558	0.5219 ^b
D12	0.5051 ^a	187.0052	0.6196 ^b
D13	1.2018 ^b	180.0345	0.5887 ^a
Ratio	^a (8/13), ^b (5/13)	^a (0/13), ^b (0/13)	^a (5/13), ^b (8/13)

^a The lowest time-consumption.

^b The middle time-consumption.

In addition to the comparison of average data recovery rates, we also compare the average time-consumption of NLS-CP, NLS-MI, and NLS-SM. The results are shown in Table 3. From Table 3, we can find the following conclusions:

- Although the average data recovery rate of NLS-CP is smaller than that of NLS-SM for D4, D6, D9, D10, and D13, average time-consumption of NLS-SM is approximately 360 times, 121 times, 370 times, 362 times, and 150 times higher than that of NLS-CP, respectively.
- For D4, D5, D6, D7, D8, D10, D11, and D12, NLS-CP has the smallest average time-consumption compared to NLS-MI and NLS-SM.
- Generally, compared with NLS-SM and NLS-MI, NLS-CP can better balance the efficiency and accuracy of noise load restoration.

5.4.3. Performance comparison of load forecasting algorithms

Although noise load detection and restoration can be a good way to improve the quality of data needed for load forecasting model, load type, season, and holiday have a direct impact on the precision of load forecasting model. Consequently, it is difficult to predict different types of loads with load forecasting model for a single type of load data. To test the impact of different types of load fusion on load forecasting, in our experiments, we used MLF-CC to perform multi-type load fusion, and obtained the experimental datasets shown in Table 4. Then, for these experimental datasets after multi-type load data fusion, load forecasting training models are firstly constructed based on LFR-MF, LR, XGBoost, SVR, DRN_STLF and APLF, and the comparison between

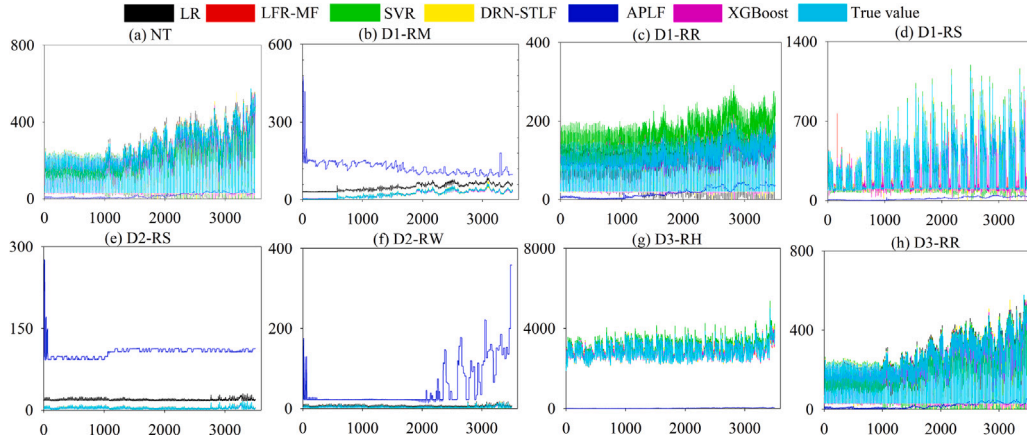


Fig. 11. Comparison of forecasting and true curves between different load forecasting models.

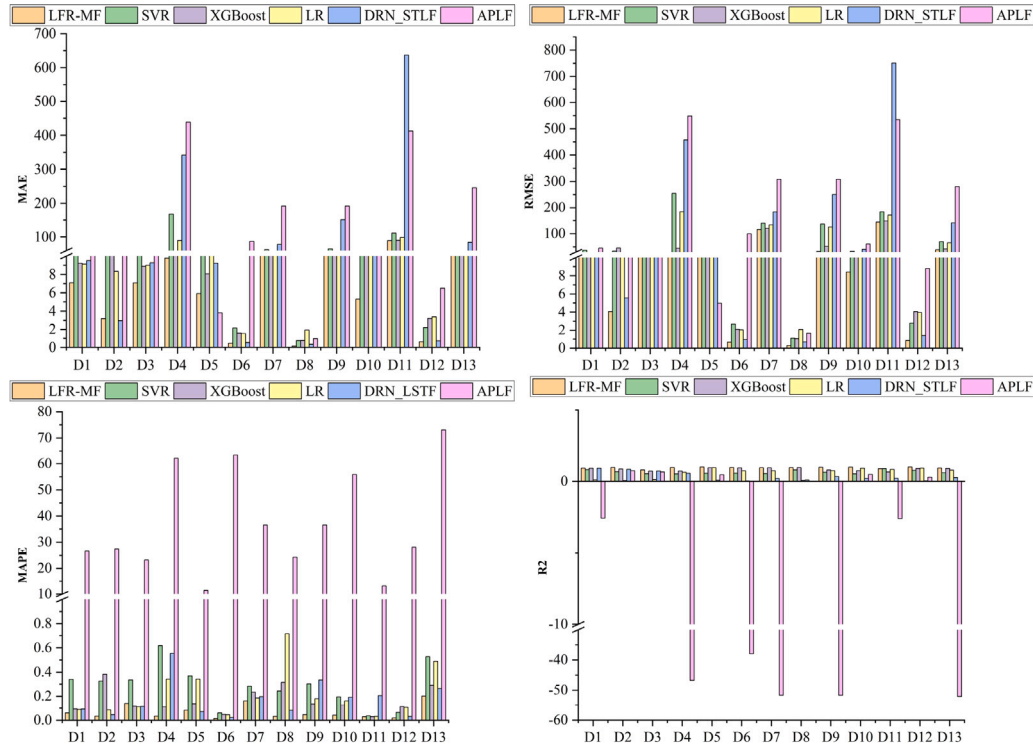


Fig. 12. Comparison of RMSE, MAE, MAPE and R^2 between different load forecasting models.

the load forecasting values and the real values of the above six models is given in Fig. 11. Fig. 12 compares RMSE, MAE, MAPE, and R^2 of all load forecasting models for all original datasets shown in Table 1.

From these Figures, we have the following conclusions:

- The prediction performance of LFR-MF is superior to SVR, XGBoost, LR, DRN-STLF, and APLF.
- From Fig. 12, by using the load forecasting model for the fused load datasets to forecast the original 13 load datasets, respectively, the experimental results indicate that MAE, RMSE, and MAPE of LFR-MF outperform the SVR, XGBoost, LR, DRN-STLF and APLF.
- Specifically, MAE of LFR-MF is higher than that of DRN-STLF and APLF by about 7.43% and 35.68% for D2 and D5 datasets. Besides, MAE of LFR-MF is better than all comparison models.
- Except for D5 dataset, LFR-MF outperforms all comparison models in terms of RMSE.

- Except for D3 and D5 datasets, MAPE of LFR-MF is significantly better than that of SVR, XGBoost, LR, DRN-STLF and APLF on the other 11 datasets.
- Except for D8 and D11 datasets, R^2 of LFR-MF outperforms all comparison models.

Remarks. In conclusion, the above experimental results show that compared with the SVR, XGBoost, LR, DRN-STLF and APLF models, the LFR-MF model proposed in this paper has better predictive ability even if the user's electricity consumption behavior and electricity consumption are quite different. This also manifests that the load forecasting model built on the multi-type load data after fusion can not only achieve higher forecasting performance for each type of load data before fusion, but also avoid building load forecasting model for each type of load data in order to save a lot of time-consumption.

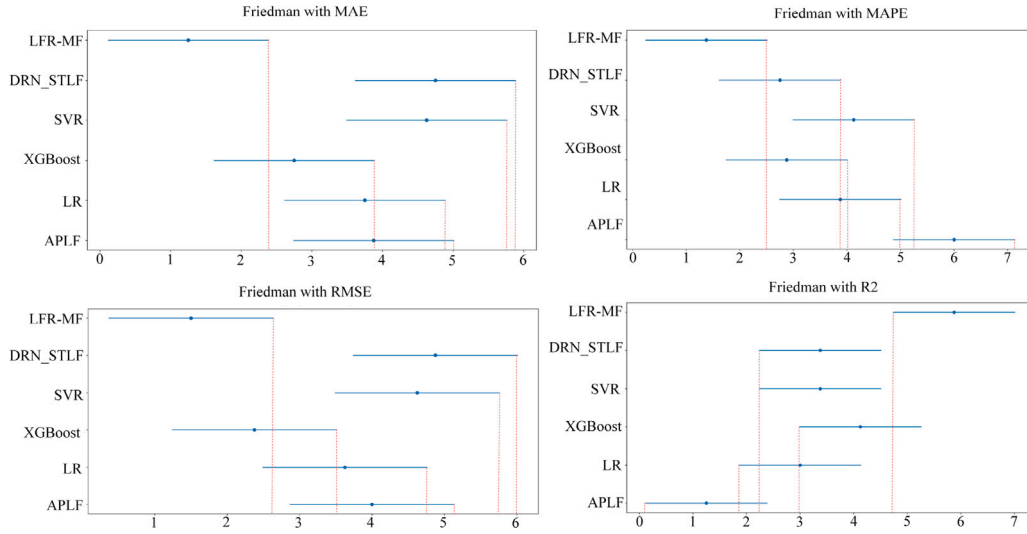


Fig. 13. Comparison of test results of six load forecasting algorithms based on Nemenyi post-hoc test.

Table 4

Experimental data after multi-type load data fusion.

Data source	Data name after fusion	#Instance after fusion	#Feature after fusion
Nantong-Resident	NT	600	96
Nantong-Commerce			
Nantong-Mixed			
District1- Market	D1-RM	584	24
District1-Resident			
District1- Resident	D1-RR	584	24
District1-Restaurant			
District1- Resident	D1-RS	584	24
District1-School			
District2-Resident	D2-RS	584	24
District2-School			
District2-Resident	D2-RW	584	24
District2-Warehouse			
District3-Resident	D3-RH	584	24
District3-Hospital			
District3-Resident	D3-RR	584	24
District3-Restaurant			

5.5. Friedman and Nimenyi test

To better test the authenticity and credibility of the experimental results in this paper, we verify the experimental results of all load forecasting algorithms based on the Friedman test. First, we calculate the test results of each algorithm on each dataset, then, rank the algorithm performance on each dataset, and finally, use the Friedman test to determine whether these load forecasting algorithms have performance differences.

There are N experimental datasets and k comparison algorithms, and let denote the average ordinal value of the i th algorithm. In the experiment, we set the variable Γ_F to obey the F distribution with $k - 1$ and $(k-1)(N - 1)$ degrees of freedom. For the datasets in Table 4, this paper conducts Friedman test and Nemenyi post-hoc test on the prediction results of LFR-MF, LR, XGBoost, SVR, DRN_STLF and APLF. The null hypotheses (H_0) and opposite hypotheses (H_1) of the Friedman test on the experimental data are denoted as follows.

- H_0 : There is no significant difference in the performance of multiple algorithms on multiple datasets;
- H_1 : There are obvious differences in the performance of multiple algorithms on multiple datasets.

Table 5

Friedman test results.

Index	Test statistic value	p-value
RMSE	23.728	0.00127
MAE	27.485	0.00027
MAPE	14.626	0.0411
R^2	16.157	0.02372

If the difference of average ordinal value between any two algorithms exceeds the critical value range, the following inequality holds.

$$(r_i - r_j) > q_\alpha \sqrt{\frac{k(k+1)}{6N}} \quad (5)$$

where q_α is the test critical value. Then we reject the null hypothesis H_0 .

If the test result rejects H_0 , it means that the performance of the k algorithms is significantly different, and then the performance of the k algorithms is further distinguished based on the Nemenyi post-hoc test.

The experimental results in Table 5 demonstrate that p-values corresponding to RMSE, MAE, MAPE and R^2 are less than 0.05, respectively. Then we reject the null hypothesis H_0 . In other words, the performance of each load forecasting algorithm is significantly different, and Nemenyi post-hoc test is required to further distinguish each algorithm. Fig. 13 shows the test results of six load forecasting algorithms based on Nemenyi post-hoc test for different evaluation metrics.

As can be seen in Fig. 13, our proposed LFR-MF algorithm has no overlapping regions with DRN_STLF, SVR and APLF in terms of MAE, RMSE and R^2 , which indicates that the LFR-MF is significantly different from the above three models. And compared with the LR, the LFR-MF has only a few overlapping regions with it on the RMSE. Although the results in Fig. 13 show that there is no significant difference in the performance of the LFR-MF compared with the XGBoost model. However, judging from the forecasting results in Fig. 11 and Fig. 12, it still outperforms XGBoost.

6. Conclusions

In this paper, we studied a new LF problem with multi-type and noisy load data, where the data matrices convey disparate variation patterns corresponding to their power consumption endeavours in which noisy entries abound due to unexpected events. A novel MLF-RFDC algorithm was proposed to deal with the multi-type and noisy load data, with its key idea being two-fold. First, noise detection and restoration

are carried out in each data matrix independently, thereby respecting their unique variation patterns. Second, the noisy-free data matrices are harmonized with random forest in a bagging and bootstrapping fashion, leading to superior model forecasting performance. A theoretical analysis was conducted on our proposed algorithm, and the results promised a tighter generalization error bound of bagging and bootstrapping over an independent LF model for multi-type load data matrices. Extensive experiments were carried out on ten benchmark datasets and three real-world load datasets and the empirical results evidenced the superior performance of our proposed MLF-RFDC over 11 competitors in terms of load noise detection, load noise restoration, and load forecasting accuracy.

In this paper, our algorithm does not account for the influence of meteorological and social factors in load forecasting. Moving forward, our model aims to enhance load forecasting accuracy by developing a new model that integrates multi-source data, including meteorological, social, and human behavior data.

CRedit authorship contribution statement

Song Deng: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Xia Dong:** Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Li Tao:** Writing – review & editing, Methodology, Data curation. **Junjie Wang:** Software, Formal analysis, Data curation. **Yi He:** Writing – review & editing, Supervision, Project administration. **Dong Yue:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Proof of Theorem 1

Proof. Due to $GE = P_{X,Y}(m_r(X,Y) < 0) = P_{X,Y}(m_r(X,Y) - E_{X,Y}(m_r(X,Y)) < -E_{X,Y}(m_r(X,Y)))$, according to Chebyshev's inequality, we have the following inequality holds

$$GE \leq \frac{Var(m_r(X,Y))}{(E_{X,Y}(m_r(X,Y)))^2} = \frac{Var(m_r(X,Y))}{s^2}. \quad (A.1)$$

Where $s = E_{X,Y}(m_r(X,Y))$. Let $\hat{j}(X,Y) = \arg \max_{j \neq Y} P_{\Theta}(T(X,\Theta) = j)$, then Eq. (2) can be changed to

$$m_r(X,Y) = P_{\Theta}(T(X,\Theta) = Y) - P_{\Theta}(T(X,\Theta) = \hat{j}(X,Y)) = E_{\Theta}(I(T(X,\Theta) = Y) - I(T(X,\Theta) = \hat{j}(X,Y))). \quad (A.2)$$

For a vector Θ , we have $E(r_{m_r}(\Theta, X, Y)) = m_r(X, Y)$. Since Θ and Θ' are two independent and identically distributed random vectors, for any function f , the following equation holds

$$(E(f(\Theta)))^2 = E_{\Theta}(f(\Theta)) \cdot E_{\Theta'}(f(\Theta')) = E_{\Theta, \Theta'}(f(\Theta) \cdot f(\Theta')). \quad (A.3)$$

Thus, $(m_r(X, Y))^2 = (E(r_{m_r}(\Theta, X, Y)))^2 = E_{\Theta, \Theta'}(r_{m_r}(\Theta, X, Y) \cdot r_{m_r}(\Theta', X, Y))$, then we have

$$\begin{aligned} Var(m_r(X, Y)) &= cov_{X,Y}(E(r_{m_r}(\Theta, X, Y)), \\ E(r_{m_r}(\Theta, X, Y))) &= E_{\Theta, \Theta'}(cov_{X,Y}(r_{m_r}(\Theta, X, Y), \\ r_{m_r}(\Theta', X, Y))) &= E_{\Theta, \Theta'}(\rho(\Theta, \Theta') \cdot s_d(\Theta) \cdot s_d(\Theta')) \\ &= E_{\Theta, \Theta'}(\rho(\Theta, \Theta')) \cdot E_{\Theta, \Theta'}(s_d(\Theta) \cdot s_d(\Theta')) \end{aligned} \quad (A.4)$$

Where $s_d(\Theta)$ is the standard deviation of $r_{m_r}(\Theta, X, Y)$. Then, Eq. (A.5) holds

$$\begin{aligned} Var(m_r(X, Y)) &= \bar{\rho} \cdot (E_{\Theta}(s_d(\Theta)))^2 \\ &\leq \bar{\rho}(E_{\Theta}(s_d(\Theta))^2) \\ &= \bar{\rho}(E_{\Theta}(Var(\Theta))) \end{aligned} \quad (A.5)$$

Where

$$\begin{aligned} Var(\Theta) &= Var(r_{m_r}(\Theta, X, Y)) \\ &= E_{X,Y}(r_{m_r}(\Theta, X, Y))^2 - (E_{X,Y}(r_{m_r}(\Theta, X, Y)))^2 \\ &= E_{X,Y}(r_{m_r}(\Theta, X, Y))^2 - s^2 \end{aligned} \quad (A.6)$$

And since $r_{m_r}(\Theta, X, Y) = I(T(X, \Theta) = Y) - I(T(X, \Theta) = \hat{j}(X, Y)) \leq 1$. From Eq. (A.6), we have $Var(\Theta) \leq 1 - s^2$. Then, the following inequality holds

$$E_{\Theta}(Var(\Theta)) \leq 1 - s^2. \quad (A.7)$$

We bring Eq. (A.7) into Eq. (A.5), and obtain

$$Var(m_r(X, Y)) \leq \bar{\rho}(1 - s^2). \quad (A.8)$$

We then bring Eq. (A.8) into Eq. (A.1) to obtain the following inequality holds

$$GE \leq \frac{Var(m_r(X, Y))}{s^2} \leq \frac{\bar{\rho}(1 - s^2)}{s^2}. \quad (A.9)$$

This completes the proof. $\square$

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